

Spatial rescue in biofilms: a double-edged sword

May 16, 2026

Abstract

A biofilm under antibiotic treatment has a clear spatial asymmetry: cells on the surface are exposed to the drug, while cells in the deeper interior are shielded from it. We ask whether that shielded interior makes the population more likely to evolve drug resistance and survive, a process known as evolutionary rescue. Comparing a spatially structured biofilm model against a well-mixed population of the same initial size, and sweeping the antibiotic dose along with the metabolic activity of the interior, we find that the biofilm is a double-edged sword; it rescues less often than the well-mixed reference at low dose, but more often at high dose, and only when the interior is metabolically active. We explain this pattern as a consequence of two opposing effects of the spatial structure: a supply mechanism that generates resistant mutants in the interior yet is subject to drift since it is not exposed to the drug, and a delivery mechanism that transports those mutants to the surface where they can grow and establish but is tied to antibiotic dose via the velocity at which the biofilm deteriorates. Our results suggest that the spatial structure of biofilms can both hinder and facilitate evolutionary rescue, and that the balance between these effects depends on the antibiotic dose and the metabolic activity of the interior.

1 Introduction

Antibiotic resistance is an escalating clinical problem, and biofilms (dense, surface-attached bacterial communities) carry a disproportionate share of it. Biofilms are responsible for a large fraction of chronic infections and are notoriously hard to clear with antibiotics; standard regimens that work in well-mixed culture often fail when the target population is organized as a biofilm. This has driven interest in behavior-modifying interventions, in particular compounds that induce bacterial dispersal away from biofilms toward a planktonic, or more well-mixed state. Such interventions do not kill cells; they simply change how cells are arranged in space. The hope is that disrupting biofilm spatial structure reduces the population's capacity to evolve resistance under antibiotic pressure, essentially disarming the bacteria such that they will be more easily cleared by traditional antibiotic treatments and the host immune system. Knowing whether that hope is justified requires understanding what the spatial structure of a biofilm actually does to a population's evolutionary dynamics in the first place. Does biofilm formation help the population evolve resistance, hurt it, or some combination of both?

Biofilms have a clear spatial asymmetry: cells on the outer surface are exposed to the drug, while cells in the deeper interior are shielded from it. The shielded interior is a spatial refuge, a region of the population where the antibiotic selection pressure does not reach. We ask whether

that refuge makes the population more likely to undergo *evolutionary rescue*, the outcome in which a declining population avoids extinction because a resistant mutant arises and takes over in time. Concretely: does a biofilm with a shielded interior rescue more often than a well-mixed population of the same initial size?

The naive prediction is that the refuge should help. A shielded interior lets the population persist for longer under treatment, and a longer life means more cell divisions, which means more opportunities for a resistant mutation to arise. More mutations should mean more rescue. But this prediction has at least two distinct ways it could be wrong. First, a mutation that arises in the shielded interior does nothing on its own: there is no selection pressure inside the refuge to make a resistant cell take over, so a core mutation matters only if it eventually reaches the drug-exposed surface in time to grow under selection. The refuge might generate mutations the population can never actually use. Second, the very mechanism that maintains the refuge, namely the replacement of dying surface cells by cells flowing outward from the interior, could act against a new resistant cell trying to establish on the surface by ensuring it is surrounded by a continuously refreshed wild-type majority. The same spatial structure that buys extra time may also make each individual mutation that arises less useful. Whether the biofilm is a net help or a net hindrance therefore depends on how these effects balance.

We measure that balance directly. We simulate a minimum-viable biofilm (a one-dimensional chain of cells with a drug-exposed edge and a shielded core) under antibiotic pressure, and compare its rescue probability against an otherwise identical well-mixed population. By varying the antibiotic dose and the metabolic activity of the interior independently, we ask whether the biofilm's spatial structure is uniformly helpful, uniformly harmful, or, as the results below will show, contingent on both the selection pressure and on what is happening inside the refuge.

2 The model

We model the biofilm as a one-dimensional chain of cells. Real biofilms are three-dimensional, but the central asymmetry we are after (drug-exposed surface versus shielded interior) does not need three dimensions to capture. One dimension keeps the dynamics tractable while preserving the feature that matters: cells at different positions face different antibiotic concentrations.

2.1 The chain and rates

The state of the population at any time t is an ordered list of cells indexed $0, 1, \dots, N(t) - 1$. Index 0 is the deep interior; index $N(t) - 1$ is the outer surface. Each cell has a genotype, either wild-type (WT) or resistant (R), and a position determined by its index. The total chain length $N(t)$ changes as cells are born and die.

Antibiotic exposure is a step function in position. The rightmost l cells of the chain form the **edge**, the region where antibiotic concentration is lethal to WT. All cells with index $< N - l$ form the **core**, the shielded interior. The boundary between edge and core sits at position

$$b(t) = \max(0, N(t) - l), \quad (1)$$

so as the population shrinks, the boundary retreats from the surface toward the interior. When $N(t) \leq l$ the boundary collapses to $b = 0$, and the entire population is edge; no core remains. The flat step-function profile is a stylization of the real, smooth antibiotic gradient through a biofilm.

Treating the boundary as sharp is the minimum-viable abstraction that captures the asymmetry; a smooth gradient would change quantitative numbers but not the qualitative dynamics we are after.

Each cell has a birth rate and a death rate determined by its compartment and its genotype, summarized in Table 1.

	Core ($i < b$)		Edge ($i \geq b$)	
	birth	death	birth	death
Wild-type (WT)	b_c	d_c	b_e	d_e
Resistant (R)	b_c	d_c	b_R	d_R

Table 1: Per-cell birth and death rates by compartment and genotype. The eight compartment-genotype-event combinations drive the Gillespie dynamics described in §2.2.

The edge WT birth rate $b_e = 1$ sets the time unit. The edge WT death rate d_e is the antibiotic dose, swept over values $d_e > b_e$ so WT cells in the edge are always net-dying under treatment. Resistant cells in the edge have $b_R = 1$ and $d_R = 0.6$, giving them a positive net growth rate $s_R = b_R - d_R = 0.4$ once they reach the drug-exposed surface; an R lineage that establishes in the edge grows deterministically to dominance. Cells in the core, regardless of genotype, have balanced rates $b_c = d_c$. The core has no net growth on its own, and WT and R are treated identically there because the shielded interior carries no selection. The default core rate is $b_c = 0.2$, a slow but nonzero pace; the dormant-core limit is $b_c = 0$, where the interior produces no births and no deaths at all. Comparing these two limits is one of the central experiments below.¹

2.2 Dynamics

Events fire one at a time, sampled as a continuous-time Markov chain via the Gillespie algorithm. At each step we compute the total event rate R_{tot} across all eight compartment-genotype-event combinations in Table 1, advance simulated time by an exponential draw with mean $1/R_{\text{tot}}$, and pick which event fires from a categorical distribution with weights proportional to each event's rate.

A birth event picks a parent uniformly at random within its compartment-genotype subgroup. The daughter is inserted adjacent to the parent (at the parent's position or one position to the right, each with probability $1/2$), and the chain grows by one. A death event picks a victim uniformly at random within its subgroup and removes it; the chain contracts by one.

The boundary $b(t)$ is recomputed after every event according to Eq. (1). When the population grows by one and the boundary moves outward, the cell that now sits at the new boundary position has crossed from edge into core; it is reclassified as a core cell. When the population shrinks and the boundary moves inward, the cell that now sits one position outside the new boundary has crossed from core into edge; it is reclassified as an edge cell. These reclassifications are not separate dynamical events: they are bookkeeping consequences of recomputing the boundary against a chain that has just changed length by one.

¹To prevent unbounded post-rescue growth, R births in both compartments are multiplied by a Wilson-type density factor $\max(0, 1 - N/K)$, with $K = N_0$ throughout. WT births stay density-independent. The factor is approximately 1 during the wild-type decline that we study, so it does not affect the dynamics of interest; it matters only once R has taken over.

This bookkeeping is the **conveyor belt** mechanism the rest of the writeup will refer to. As long as the core has cells in it, every edge death pulls an interior cell out to the surface, and every edge birth pushes an outer cell back into the interior. The result is that the edge holds exactly l cells while the core has anything left to supply; the population is shrinking, but the surface layer is being continuously replaced from below.

The conveyor belt produces two qualitatively different regimes of decline. While the core still has cells in it ($N(t) > l$), each edge death is replaced immediately by a core cell, so the edge stays at exactly l cells. Only the l edge cells contribute to net loss, at rate $s_e \cdot l$ per unit time, and the total population falls linearly. The duration of this buffered decline is

$$T_1 = \frac{N_0 - l}{s_e \cdot l}, \quad (2)$$

long at low dose (slow edge turnover gradually drains the core) and brief at high dose (fast edge turnover quickly burns through it). Once the core is exhausted ($N(t) \leq l$), the edge is the entire remaining population, and there is no reservoir to draw on. The remaining $\leq l$ cells decay exponentially at rate s_e , just as a well-mixed population would. The transition between these two regimes is sharp on the simulation timescale: it happens at the moment the last core cell is converted to edge. After that point, the dynamics on the chain are indistinguishable from those of a one-compartment well-mixed model with $N = l$.

2.3 Mutations, rescue, and initial condition

Mutations are replication-linked: at each WT birth event, with probability μ , the daughter is an R cell instead of WT; otherwise the daughter inherits the parent's genotype. Mutations are irreversible (no back-mutation from R to WT), so once a lineage is R, all of its descendants are R. Because mutations ride on births, they appear at specific spatial positions on the chain (the position of the dividing parent); a mutation born deep in the core enters the world at the core's spatial location and must be physically transported to the edge to come under selection.

Every R lineage that ever appears is tagged with a unique identifier when its founder is born, and that identifier propagates to all descendants. This lets us follow each mutation individually: where it was born (core or edge), when it appeared, how its descendant population grew, whether it ever reached the edge, and whether it contributed to rescue. The per-lineage records produced by this tracking are the basis for most of the analyses in later sections.

A run is declared rescued when the edge contains at least $r_{\text{thr}} = 10$ resistant cells simultaneously. The choice of $r_{\text{thr}} = 10$ is high enough to be well past the establishment bottleneck (the regime where a small R lineage might still go extinct by chance) but small enough that rescue is declared promptly after takeover begins. Once an R lineage is past establishment in the edge, where $s_R > 0$, it grows deterministically to dominance. Every simulation terminates naturally on one of two events: rescue ($\geq r_{\text{thr}}$ R cells in the edge) or extinction ($N = 0$). There are no artificial time or generation caps. Both endpoints are guaranteed by the dynamics: either a successful resistant lineage establishes in time, or the wild-type population is driven to zero. The model cannot stall in between.

Every simulation starts at $t = 0$ with $N = N_0$ wild-type cells and no resistant cells. Before $t = 0$ the population is at quasi-steady state, with balanced birth and death rates in both compartments ($b_e = d_e = 1$ pre-treatment in the edge, $b_c = d_c$ in the core), so the biofilm neither grows nor shrinks. At $t = 0$ the antibiotic switches on: the edge WT death rate jumps from $b_e = 1$ to its

treatment value $d_e > 1$, and the population begins to decline. What we simulate is therefore the response of an already-mature biofilm to a step-function antibiotic challenge. Real treatments are gradual; this abstraction strips out the loading dynamics so that the rescue dynamics, the only thing we are studying, are not entangled with growth-phase or pharmacokinetic transients.

2.4 The well-mixed reference and parameters

To say whether the biofilm helps or hurts, we need a baseline. Our baseline is a well-mixed population of the same initial size N_0 , governed by the edge rates alone: every cell is drug-exposed, none are shielded. This is the standard setting for evolutionary rescue (Wilson et al., 2017), and it is the natural null model, the same biofilm with all spatial structure removed. Operationally, the well-mixed reference is the chain model with $l \geq N_0$: when the edge is as wide as the entire population, no cell is ever in the core, the conveyor belt never engages, and the dynamics reduce to a one-compartment birth-death process on scalar counts. We also maintain an independent scalar-count implementation as a cross-check; the two agree statistically within sampling error, which serves as an internal consistency test on the chain model.

Unless stated otherwise, all simulations use the parameters in Table 2. Time is measured in edge generations ($b_e = 1$). The dose sweep covers $d_e \in \{1.05, 1.1, 1.2, 1.4, 1.6, 1.9, 2.3, 2.8\}$. The core-activity sweep covers $b_c \in \{0, 0.05, 0.1, 0.2, 0.4, 0.8\}$. The edge-width sweep covers $l \in \{25, 50, 100, 200, 400\}$. Every parameter combination is run with 5,000 independent seeds; the well-mixed reference is run at the same dose points with the same seed count.

Parameter	Value	Meaning
N_0	1000	Initial population size
l	100	Edge width (cells exposed to drug)
b_e	1.0	WT birth rate in edge (sets time unit)
d_e	dose (swept)	WT death rate in edge; super-MIC is $d_e > 1$
b_R	1.0	R birth rate in edge
d_R	0.6	R death rate in edge ($s_R = 0.4$)
$b_c = d_c$	0.2 (swept)	Core birth/death rate (0 = dormant)
μ	10^{-4}	Mutation probability per birth
r_{thr}	10	Resistant edge cells required to declare rescue
K	N_0	Density cap for R births (see footnote, §2.1)

Table 2: Default model parameters. The swept variables are dose (d_e), core activity (b_c), and edge width (l).

3 The double-edged sword

Section 1 closed with a prediction (the refuge should help) and two distinct ways the prediction could be wrong. The refuge could be useless on its own, with the extra lifetime never translating into extra rescue. Or the refuge could be helpful only under specific conditions, say only when the antibiotic dose is high or only when the interior is metabolically active, so that the average effect across regimes hides a sharp contingency. These are different failure modes, and they imply different things about what spatial structure does for a population trying to evolve resistance. We want to distinguish them in a single experiment.

The cleanest way to do that is to vary two knobs independently. The first knob is the antibiotic dose, which we sweep between a value just above the minimum inhibitory concentration ($d_e = 1.05$) and a value well above it ($d_e = 2.8$). The second is the metabolic activity of the interior, which we set to two values: $b_c = 0$ (dormant; the interior is a passive refuge with no births and no deaths) and $b_c = 0.8$ (active; the interior is dividing at a substantial fraction of the edge rate). At each of the four combinations we run 5,000 independent simulations of the biofilm and 5,000 of the well-mixed reference, and compute the fraction that rescue.

Each of the two knobs isolates a different question. The dormant-core column ($b_c = 0$) asks whether the geometric refuge by itself matters: with no births in the interior, the only thing the biofilm gains over the well-mixed reference is extra lifetime; does that alone help? The active-core column ($b_c = 0.8$) asks whether what happens inside the refuge matters: with the interior dividing, the biofilm has extra cell divisions on top of the lifetime extension; do those help? And running both at low and high dose asks whether the answers depend on the selection pressure that is forcing rescue in the first place. Four cells of a 2×2 design, three live hypotheses tested at once.

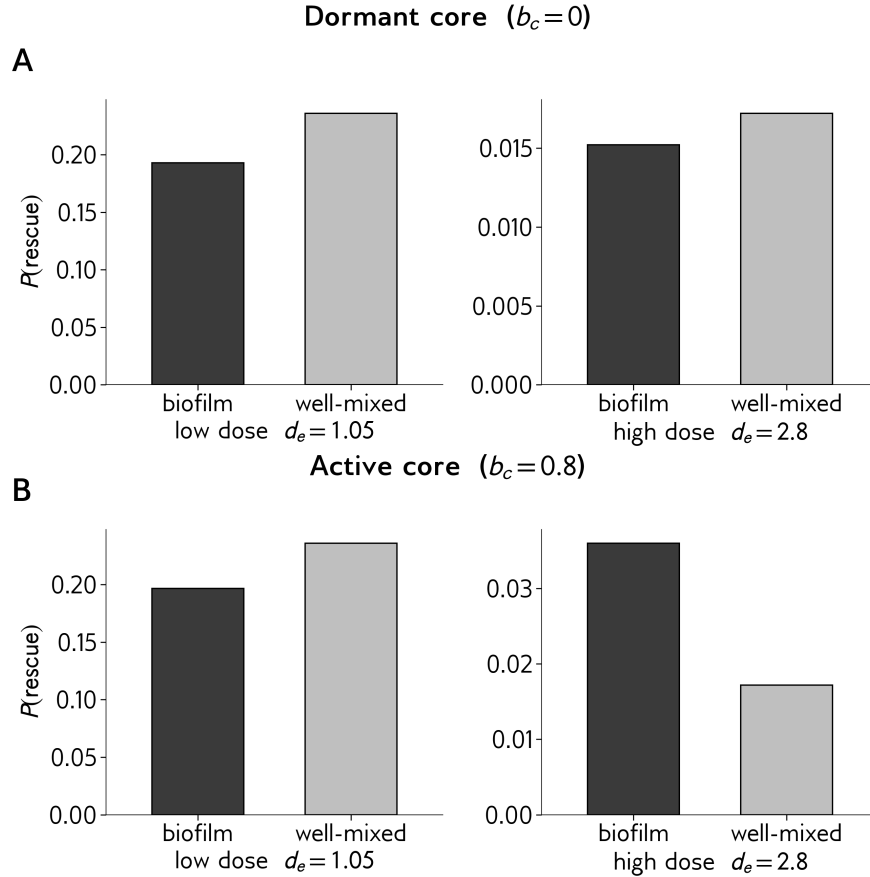


Figure 1: Rescue probability for the biofilm (dark) versus the well-mixed reference (light), at low dose ($d_e = 1.05$, left columns) and high dose ($d_e = 2.8$, right columns). Panel A: dormant core ($b_c = 0$). Panel B: active core ($b_c = 0.8$). Each subplot uses its own y-axis scale. *Parameters:* $N_0 = 1000$, $l = 100$, $\mu = 10^{-4}$; each bar averages over 5,000 independent simulation runs.

The result, shown in Figure 1, reduces to two clean observations rather than four panel-by-

panel readings. First, the dormant-core column is uniformly low: at both low dose and high dose, the biofilm with a passive interior rescues less often than the well-mixed reference, with the dark bars sitting strictly below the light bars in Panel A. The geometric refuge by itself (extra lifetime, no extra divisions) does not give the biofilm a rescue advantage at any dose we tested. Whatever the shielded interior is buying the population in lifetime, it is not being converted into extra rescue probability when no division happens inside.

Second, within the active-core column, the dose decides. With the interior dividing at $b_c = 0.8$ (Panel B), the low-dose comparison still favors the well-mixed reference; but the high-dose comparison inverts, with the biofilm rescuing roughly twice as often as the well-mixed reference at $d_e = 2.8$. The same biofilm, with the same parameters, the same conveyor belt, and the same active core, is a net hindrance at low dose and a net help at high dose.

Of the four cells of the 2×2 , only one (high dose paired with an active interior) flips the comparison in the biofilm's favor. A dormant interior never helps regardless of dose; a high dose alone never helps if the interior cannot divide. The biofilm's advantage emerges only at the intersection of the two conditions; neither is sufficient on its own.

This is the *double-edged sword*. The same biofilm, the same spatial structure, the same conveyor belt, operating at different doses, with the comparison flipping from net-harmful to net-helpful as the dose crosses some threshold between $d_e = 1.05$ and $d_e = 2.8$. Spatial structure is neither friend nor foe; what it is depends on the regime. The rest of this writeup is about why.